

1 Summary

Model (Run-name): model3 (test)

Iteration: 10

TAC: 63882.04411793158

Took: 185.49504780769348s

Total Time: 648.3266460895538s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

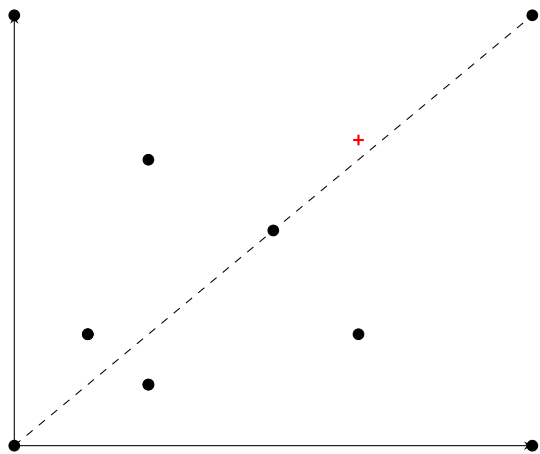
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{3,5,1}$	22.3	$q_{3,5,1}$	1544.29000000000002
$z_{2,1,1}$	1.0	$dt_{3,5,2}$	29.05608108108188	$q_{4,3,1}$	1539.72
$z_{2,2,1}$	1.0	$dt_{4,3,1}$	50.0	$q_{5,4,1}$	1634.85
$z_{3,5,1}$	1.0	$dt_{4,3,2}$	111.1	$q_{cu,1}$	357.41000000000011
$z_{4,3,1}$	1.0	$dt_{5,4,1}$	22.2	$q_{cu,3}$	839.99000000000033
$z_{5,4,1}$	1.0	$dt_{5,4,2}$	24.33559322033899	$q_{cu,5}$	724.55999999999988
$z_{cu,1}$	1.0	$dt_{cu,1}$	93.91477272727286	$t_{1,1}^{(H)}$	160.0
$z_{cu,3}$	1.0	$dt_{cu,3}$	82.3560810810813	$t_{1,2}^{(H)}$	160.0
$z_{cu,5}$	1.0	$dt_{cu,5}$	66.53559322033891	$t_{1,3}^{(H)}$	133.91477272727286
$A_{1,1,2}^{\beta}$	2.000219995065596	$f_{1,1,2}^H$	8.8	$t_{2,1}^{(H)}$	248.9
$A_{2,1,1}^{\beta}$	2.958362968240719	$f_{2,1,1}^H$	4.774527452745286	$t_{2,2}^{(H)}$	137.8
$A_{2,2,1}^{\beta}$	7.8388326972047775	$f_{2,2,1}^H$	5.8254725472547255	$t_{2,3}^{(H)}$	137.8
$A_{3,5,1}^{\beta}$	12.923002798326563	$f_{3,5,1}^H$	9.585909373060213	$t_{3,1}^{(H)}$	226.7
$A_{4,3,1}^{\beta}$	6.676114802576327	$f_{4,3,1}^H$	12.6	$t_{3,2}^{(H)}$	122.35608108108121
$A_{5,4,1}^{\beta}$	14.14358715764228	$f_{5,4,1}^H$	12.26444111027757	$t_{3,3}^{(H)}$	122.3560810810813
$A_{cu,1}^{\beta}$	2.044334101391427	$f_{1,1,2}^C$	2.2954999999999877	$t_{4,1}^{(H)}$	271.1
$A_{cu,3}^{\beta}$	5.422960781391782	$f_{2,1,1}^C$	7.6	$t_{4,2}^{(H)}$	148.9
$A_{cu,5}^{\beta}$	5.324393725088925	$f_{2,2,1}^C$	6.1	$t_{4,3}^{(H)}$	148.9
$A_{1,1,2}$	3.460657649345137	$f_{3,5,1}^C$	13.9	$t_{5,1}^{(H)}$	198.9
$A_{2,1,1}$	9.439495705511714	$f_{4,3,1}^C$	8.4	$t_{5,2}^{(H)}$	106.53559322033891
$A_{2,2,1}$	30.932782824910646	$f_{5,4,1}^C$	17.3	$t_{5,3}^{(H)}$	106.53559322033891
$A_{3,5,1}$	71.16597394333074	$RecLMTD_{1,1,2}$	0.012814458732055667	$t_{1,1}^{(C)}$	160.0
$A_{4,3,1}$	23.670439786681715	$RecLMTD_{2,1,1}$	0.015125971061711659	$t_{1,2}^{(C)}$	90.20394736842088
$A_{5,4,1}$	82.7196877037195	$RecLMTD_{2,2,1}$	0.04062493688474228	$t_{1,3}^{(C)}$	60.0
$A_{cu,1}$	4.5836803758229046	$RecLMTD_{3,5,1}$	0.039170802020235526	$t_{2,1}^{(C)}$	221.7
$A_{cu,3}$	16.73892021814917	$RecLMTD_{4,3,1}$	0.013067228988828776	$t_{2,2}^{(C)}$	115.6
$A_{cu,5}$	16.23492858720698	$RecLMTD_{5,4,1}$	0.04300806468370896	$t_{2,3}^{(C)}$	115.6
$dt_{1,1,2}$	69.79605263157912	$RecLMTD_{cu,1}$	0.01782710325509921	$t_{3,1}^{(C)}$	221.1
$dt_{1,1,3}$	73.91477272727288	$RecLMTD_{cu,3}$	0.016938394725445186	$t_{3,2}^{(C)}$	37.8
$dt_{2,1,1}$	88.9	$RecLMTD_{cu,5}$	0.019045612922499865	$t_{3,3}^{(C)}$	37.8
$dt_{2,1,2}$	47.596052631579	$q_{1,1,2}$	229.55	$t_{4,1}^{(C)}$	176.7
$dt_{2,2,1}$	27.2	$q_{2,1,1}$	530.45	$t_{4,2}^{(C)}$	82.2
$dt_{2,2,2}$	22.2	$q_{2,2,1}$	647.2099999999999	$t_{4,3}^{(C)}$	82.2

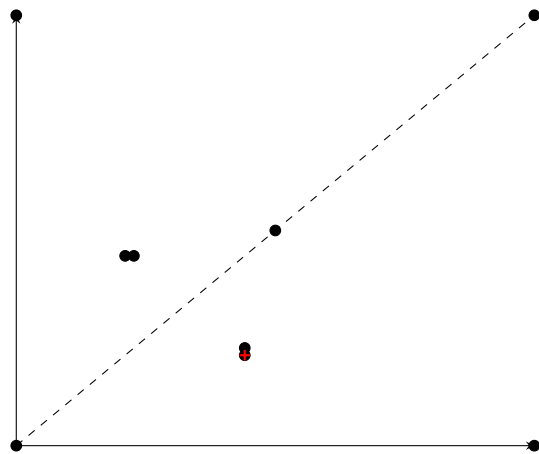
Variable	Value	Variable	Value
$t_{5,1}^{(C)}$	204.4	$b_{1,1,2}^{C,\text{out}}$	367.2799999999998
$t_{5,2}^{(C)}$	93.3	$b_{2,1,1}^{C,\text{out}}$	1216.0
$t_{5,3}^{(C)}$	93.3	$b_{2,2,1}^{C,\text{out}}$	1352.37
$t_{1,1,2}^H$	133.91477272727286	$b_{3,5,1}^{C,\text{out}}$	2841.16
$t_{2,1,1}^H$	137.8	$b_{4,3,1}^{C,\text{out}}$	1857.24
$t_{2,2,1}^H$	142.37311557276922	$b_{5,4,1}^{C,\text{out}}$	3056.91
$t_{3,5,1}^H$	72.64443425981062		
$t_{4,3,1}^H$	148.9		
$t_{5,4,1}^H$	65.6		
$t_{1,1,2}^C$	148.9083888138303		
$t_{2,1,1}^C$	160.0		
$t_{2,2,1}^C$	221.7		
$t_{3,5,1}^C$	204.4		
$t_{4,3,1}^C$	221.1		
$t_{5,4,1}^C$	176.7		
$b_{1,1,2}^{H,\text{in}}$	1408.0		
$b_{2,1,1}^{H,\text{in}}$	1188.3798829883017		
$b_{2,2,1}^{H,\text{in}}$	1449.9601170117012		
$b_{3,5,1}^{H,\text{in}}$	2173.12565487275		
$b_{4,3,1}^{H,\text{in}}$	3415.8599999999997		
$b_{5,4,1}^{H,\text{in}}$	2439.397336834209		
$b_{1,1,2}^{H,\text{out}}$	1178.45		
$b_{2,1,1}^{H,\text{out}}$	657.9298829883005		
$b_{2,2,1}^{H,\text{out}}$	802.7501170117013		
$b_{3,5,1}^{H,\text{out}}$	628.83565487275		
$b_{4,3,1}^{H,\text{out}}$	1876.1399999999999		
$b_{5,4,1}^{H,\text{out}}$	804.5473368342082		
$b_{1,1,2}^{C,\text{in}}$	137.72999999999925		
$b_{2,1,1}^{C,\text{in}}$	685.55		
$b_{2,2,1}^{C,\text{in}}$	705.16		
$b_{3,5,1}^{C,\text{in}}$	1296.8699999999997		
$b_{4,3,1}^{C,\text{in}}$	317.52		
$b_{5,4,1}^{C,\text{in}}$	1422.0599999999999		

3 *RecLMTD* Tangent Points

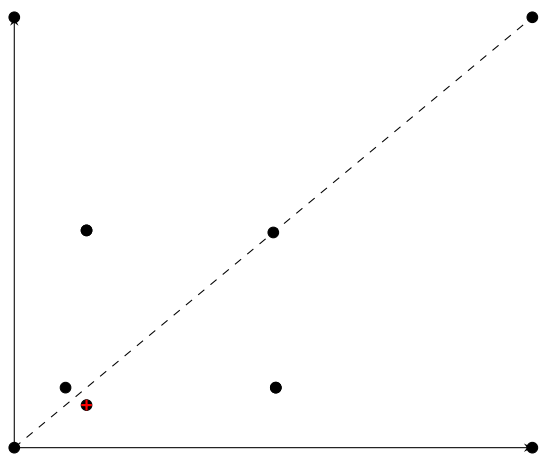
- (1, 1, 2)
 - Value: 0.012814
 - Error: 0.001106189580598
 - Relative error: 0.079463941316039
- (2, 1, 1)
 - Value: 0.015126
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000001
- (2, 2, 1)
 - Value: 0.040625
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000032
- (3, 5, 1)
 - Value: 0.039171
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000006
- (4, 3, 1)
 - Value: 0.013067
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (5, 4, 1)
 - Value: 0.043008
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000001
- $cu, 1$
 - Value: 0.017827
 - Error: 0.005393733859502
 - Relative error: -0.433811116511316
- $cu, 3$
 - Value: 0.016938
 - Error: 0.000000077787154
 - Relative error: 0.000004592335799
- $cu, 5$
 - Value: 0.019046
 - Error: 0.000000374191260
 - Relative error: 0.000019646724400



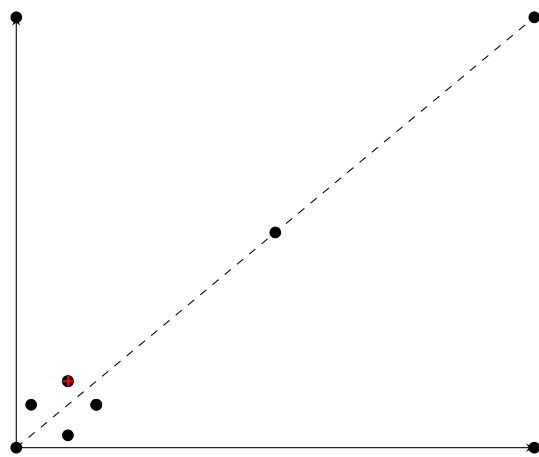
(a) *RecLMTD* - (1, 1, 2)



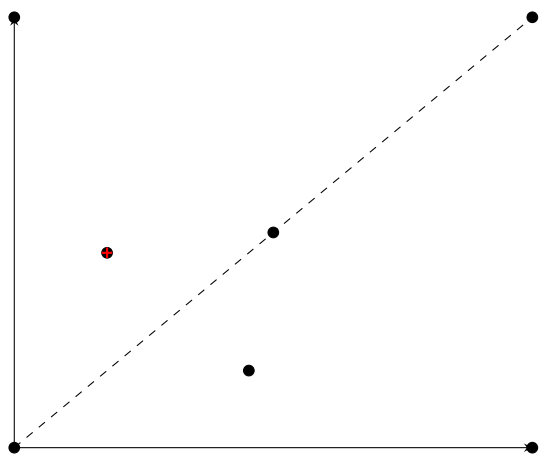
(b) *RecLMTD* - (2, 1, 1)



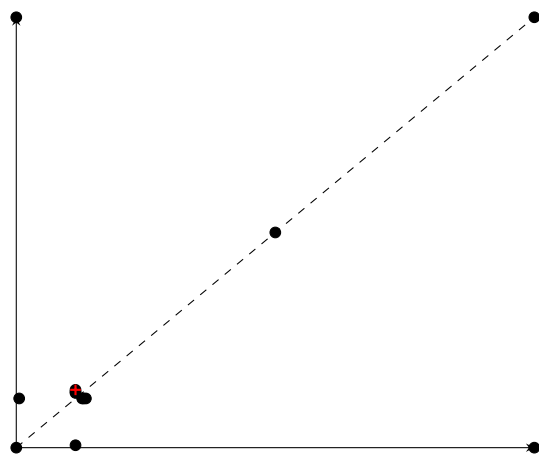
(c) *RecLMTD* - (2, 2, 1)



(d) *RecLMTD* - (3, 5, 1)



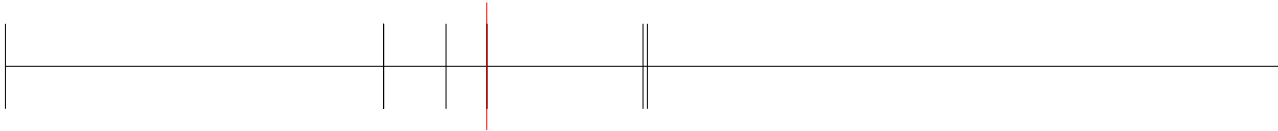
(e) *RecLMTD* - (4, 3, 1)



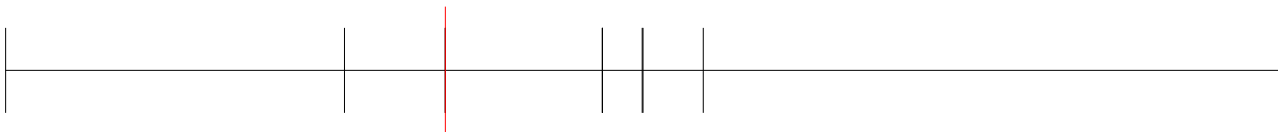
(f) *RecLMTD* - (5, 4, 1)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - $cu, 3$



(c) *RecLMTD* - $cu, 5$

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 1188.3798829883017
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.9601170117012
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2173.12565487275
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 1)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 4, 1)
 - Value: 2439.397336834209
 - Absolute error: 4.547473508864641e-13
 - Relative error: 1.8641790905479316e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 1178.45
 - Absolute error: 1.1368683772161603e-12
 - Relative error: -9.647149876669856e-16
- (2, 1, 1)
 - Value: 657.9298829883005
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.7279476227049206e-16
- (2, 2, 1)

- Value: 802.7501170117013
 - Absolute error: 26.640559224590106
 - Relative error: -0.03212063987201163
- (3, 5, 1)
 - Value: 628.83565487275
 - Absolute error: 67.52730839902517
 - Relative error: -0.09697142432985878
- (4, 3, 1)
 - Value: 1876.1399999999999
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.2119227533298796e-16
- (5, 4, 1)
 - Value: 804.5473368342082
 - Absolute error: 3.410605131648481e-13
 - Relative error: -4.2391603023245075e-16

4.3 $b^{C,in}$

- (1, 1, 2)
 - Value: 137.729999999999925
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 1)
 - Value: 685.55
 - Absolute error: 1.2505552149377763e-12
 - Relative error: 1.8241633942641366e-15
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 1296.8699999999997
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.7532495581147846e-16
- (4, 3, 1)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0

- (5, 4, 1)
 - Value: 1422.0599999999999
 - Absolute error: 1.1368683772161603e-12
 - Relative error: -7.994517651970804e-16

4.4 $b^{C,out}$

- (1, 1, 2)
 - Value: 367.2799999999998
 - Absolute error: 25.46079347785235
 - Relative error: 0.07448614060311091
- (2, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2841.16
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.6005693128386436e-16
- (4, 3, 1)
 - Value: 1857.24
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 1)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0



$$(a) \ t^{(H)} - (1, 2)$$



$$(a) \ t^{(H)} - (2, 1)$$



$$(a) \ t^{(H)} - (2, 1)$$



$$(a) \ t^{(H)} - (3, 1)$$



$$(a) \ t^{(H)} - (4, 1)$$



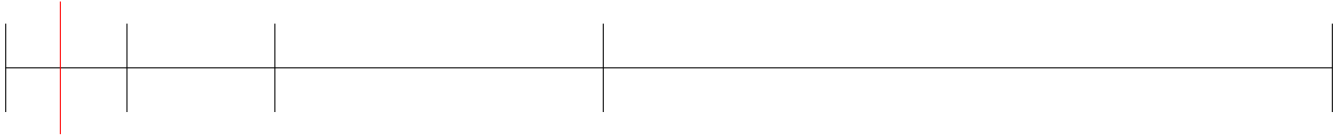
$$(a) \ t^{(H)} - (5, 1)$$



$$(a) \ t^{(H)} - (1, 1, 2)$$



$$(a) \ t^{(H)} - (2, 1, 1)$$



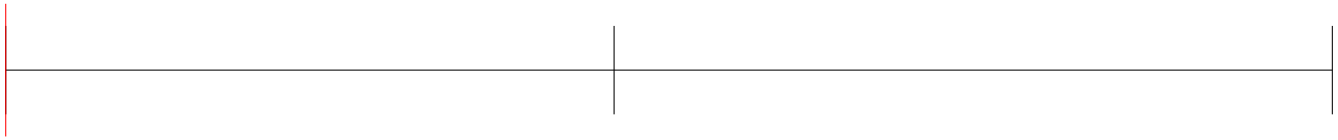
(a) $t^{(H)} - (2, 2, 1)$



(a) $t^{(H)} - (3, 5, 1)$



(a) $t^{(H)} - (4, 3, 1)$



(a) $t^{(H)} - (5, 4, 1)$



(a) $t^{(C)} - (1, 3)$



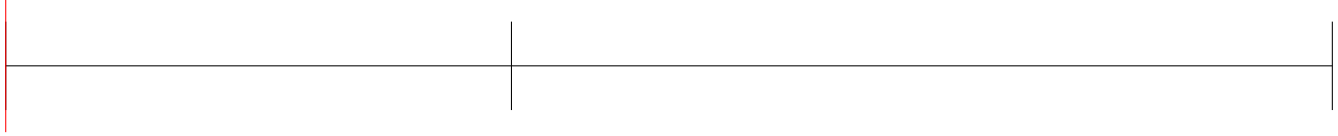
(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 2)$



(a) $t^{(C)} - (5, 2)$



(a) $t^{(C)} - (3, 2)$



(a) $t^{(C)} - (4, 2)$



(a) $t^{(C)} - (1, 1, 2)$



(a) $t^{(C)} - (2, 1, 1)$



(a) $t^{(C)} - (2, 2, 1)$



(a) $t^{(C)} - (3, 5, 1)$



(a) $t^{(C)} - (4, 3, 1)$



(a) $t^{(C)} - (5, 4, 1)$

5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 3.460658
 - Error: 0.0000000000000014
 - Relative error: 0.0000000000000004
- $(2, 1, 1)$
 - Value: 9.439496
 - Error: 0.0000000000000007
 - Relative error: -0.0000000000000001
- $(2, 2, 1)$
 - Value: 30.932783
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(3, 5, 1)$
 - Value: 71.165974
 - Error: 0.0000000000001876
 - Relative error: -0.0000000000000026
- $(4, 3, 1)$
 - Value: 23.670440
 - Error: 0.0000000000000018
 - Relative error: -0.0000000000000001
- $(5, 4, 1)$
 - Value: 82.719688
 - Error: 0.0000000000000028
 - Relative error: 0.0000000000000000

5.2 A_{CU}

- $cu, 1$
 - Value: 4.583680
 - Error: 2.912301947006541
 - Relative error: 0.388515050007116
- $cu, 3$
 - Value: 16.738920
 - Error: 0.0000000000000043
 - Relative error: -0.0000000000000003
- $cu, 5$

- Value: 16.234929
- Error: 0.000000000000643
- Relative error: 0.000000000000040



(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 1, 1)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (3, 5, 1)$



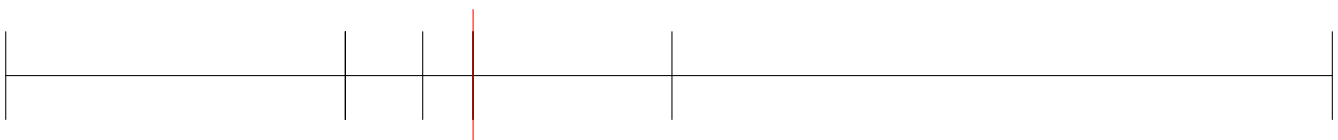
(a) $A^\beta - (4, 3, 1)$



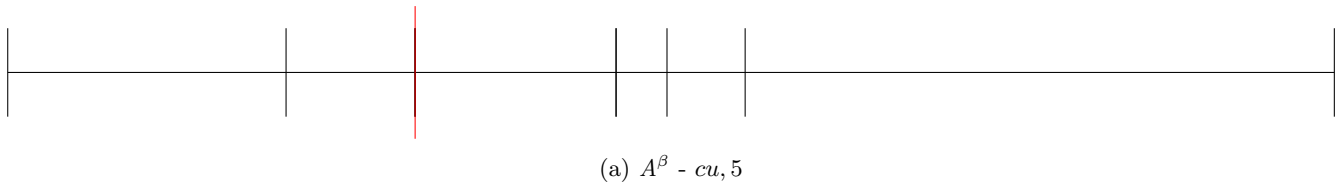
(a) $A^\beta - (5, 4, 1)$



(a) $A^\beta - cu, 1$



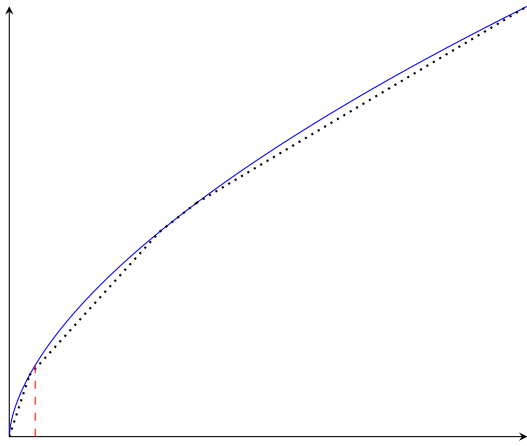
(a) $A^\beta - cu, 3$



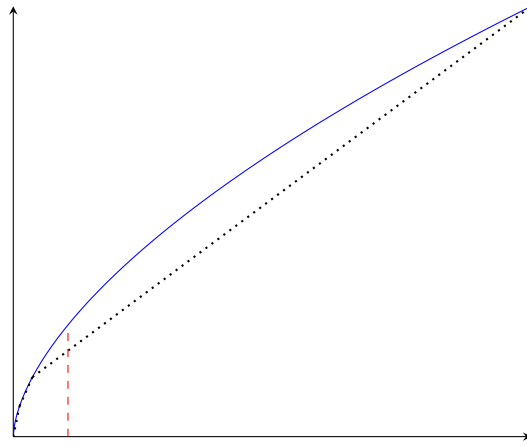
6 A^β Breakpoints

- (1, 1, 2)
 - q : 3.460658
 - $\hat{q}^\beta$: 2.000220
 - q^β : 2.106179
 - Breakpoints: [0, **2.7005882352941173**, **20.05363555217427**, 24.928068483288385, 69.05411764705883]
 - Error: 0.105958542612206
 - Relative error: 0.050308433362459
- (2, 1, 1)
 - q : 9.439496
 - $\hat{q}^\beta$: 2.958363
 - q^β : 3.845646
 - Breakpoints: [0, 1.0776943916793027, **3.3036464983028657**, **89.41176470588236**]
 - Error: 0.887282783230306
 - Relative error: 0.230724003346097
- (2, 2, 1)
 - q : 30.932783
 - $\hat{q}^\beta$: 7.838833
 - q^β : 7.838833
 - Breakpoints: [0, 1.9849079917037988, 13.879417919656465, 16.512573095835464, 18.890814020083003, **30.93278282491032**, **30.93278282491164**, 76.14235294117645]
 - Error: 0.000000000000240
 - Relative error: -0.000000000000031
- (3, 5, 1)
 - q : 71.165974
 - $\hat{q}^\beta$: 12.923003
 - q^β : 12.923003
 - Breakpoints: [0, 30.750913778350093, 44.57376487924321, **71.16597394333036**, **181.68117647058827**]
 - Error: 0.000000000000039
 - Relative error: 0.000000000000003
- (4, 3, 1)
 - q : 23.670440
 - $\hat{q}^\beta$: 6.676115
 - q^β : 6.676115
 - Breakpoints: [0, 19.921388087224845, 23.670439786681058, **23.670439786681666**, **181.14352941176475**]
 - Error: 0.000000000000003
 - Relative error: 0.000000000000000

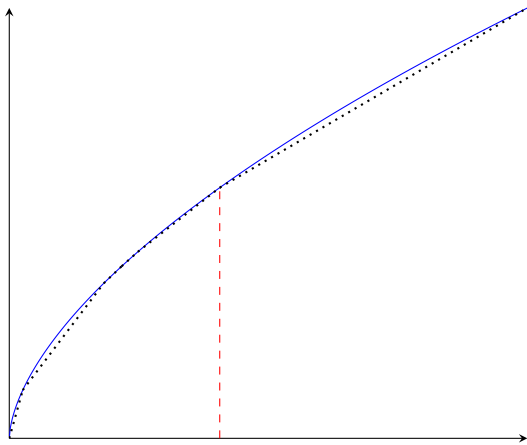
- (5, 4, 1)
 - q : 82.719688
 - $\hat{q}^\beta$: 14.143587
 - q^β : 14.143775
 - Breakpoints: [0, 7.000222767756085, 12.878068451030797, 38.682379459353704, 82.70899711080247, 192.3352941176471]
 - Error: 0.000187605921026
 - Relative error: 0.000013264204511
- $cu, 1$
 - q : 4.583680
 - $\hat{q}^\beta$: 2.044334
 - q^β : 2.493036
 - Breakpoints: [0, 7.527593053896052, 7.5275930538960525, 7.5275930538960925, 22.75735506822926]
 - Error: 0.448702162105622
 - Relative error: 0.179982204300637
- $cu, 3$
 - q : 16.738920
 - $\hat{q}^\beta$: 5.422961
 - q^β : 5.422961
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 13.643579101040302, 14.376341772690381, 16.73892021814917, 16.738997089244634, 16.738997089244645, 18.852085704732907, 128.44338870924773]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 5$
 - q : 16.234929
 - $\hat{q}^\beta$: 5.324394
 - q^β : 5.324395
 - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 16.235135805353295, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.000000816978476
 - Relative error: 0.000000153440634



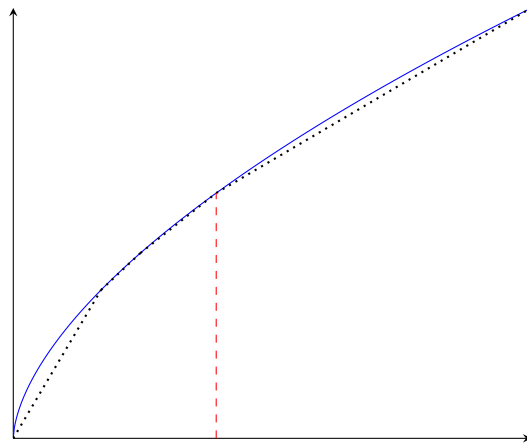
(a) $q^\beta - (1, 1, 2)$



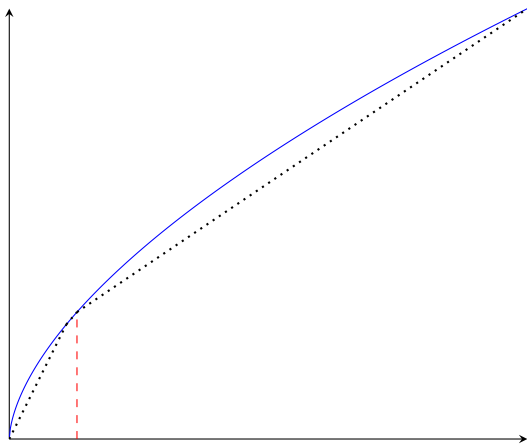
(b) $q^\beta - (2, 1, 1)$



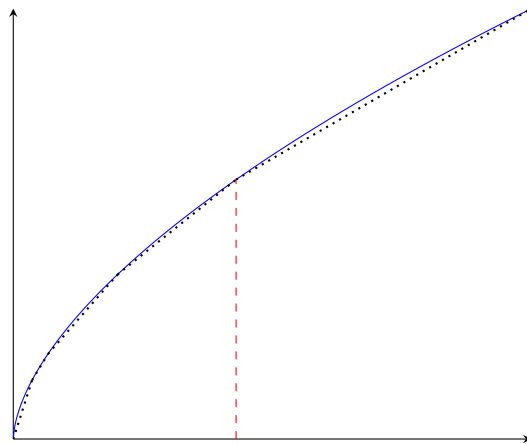
(c) $q^\beta - (2, 2, 1)$



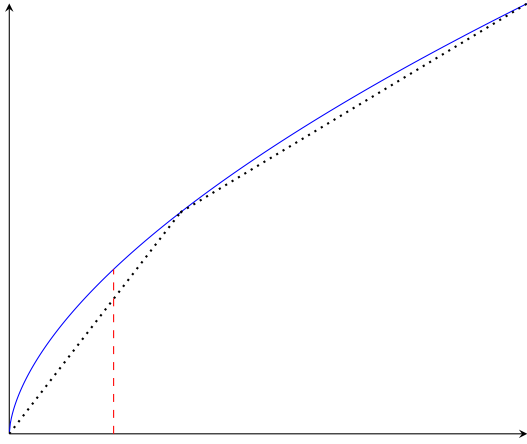
(d) $q^\beta - (3, 5, 1)$



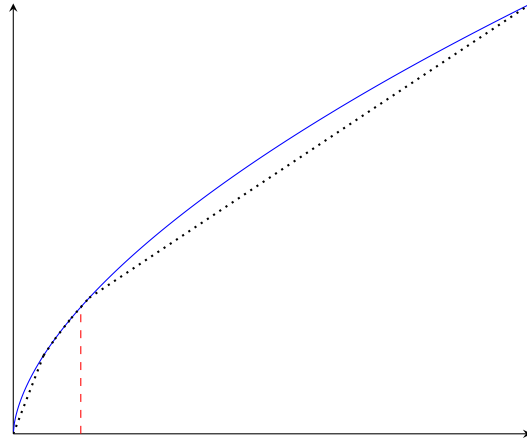
(e) $q^\beta - (4, 3, 1)$



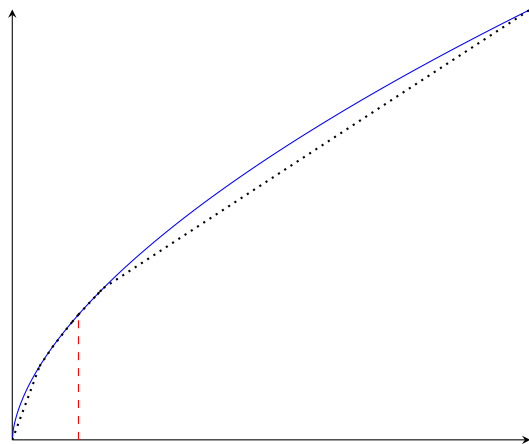
(f) $q^\beta - (5, 4, 1)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 3$



(c) $q^\beta - cu, 5$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,1}$	0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{4,5,2}$	0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,2}$	0.0	$z_{5,1,1}$	0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,1,2}$	0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,2}$	0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,2,2}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,2}$	0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,3,2}$	0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,2}$	0.0	$z_{5,4,2}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{2,1,2}$	0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,2,2}$	0.0	$z_{5,5,2}$	0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,3,1}$	0.0	$z_{cu,2}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,2}$	0.0	$z_{cu,4}$	0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,4,1}$	0.0	$z_{hu,1}$	0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,2}$	0.0	$z_{hu,2}$	0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	0.0	$z_{hu,3}$	0.0	$A_{4,3,2}^\beta$	0.0
$z_{2,5,2}$	0.0	$z_{hu,4}$	0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	0.0	$z_{hu,5}$	0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	0.0	$A_{1,2,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,2}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,4,2}^\beta$	0.0
$z_{4,1,2}$	0.0	$A_{2,1,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	0.0	$A_{2,2,2}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,3,2}$	0.0	$A_{2,3,2}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,1}$	0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,2}$	0.0	$A_{2,4,2}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,3,2}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{2,1,3}$	100.0
$A_{hu,5}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,1,1}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,2,2}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,1}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,3,2}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,4,2}$	55.6
$A_{1,4,1}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,4,3}$	94.5
$A_{1,4,2}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,2}$	10.0
$A_{1,5,2}$	0.0	$A_{5,4,2}$	0.0	$dt_{2,5,3}$	111.1
$A_{2,1,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,2,2}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,3,1}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,3}$	156.7560810810815
$A_{2,3,2}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,4,1}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	111.1
$A_{2,4,2}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	111.1
$A_{2,5,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,2}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	10.0
$A_{3,1,1}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	188.9
$A_{3,1,2}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,1}$	144.5
$A_{3,2,1}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,4,2}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,2,2}$	17.15006133296988	$dt_{3,4,3}$	144.5
$A_{3,3,1}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{3,5,3}$	133.4
$A_{3,3,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,1}$	122.2
$A_{3,4,1}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,1,2}$	69.79605263157912
$A_{3,4,2}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,1,3}$	100.0
$A_{3,5,2}$	0.0	$dt_{1,4,1}$	10.0	$dt_{4,2,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,4,2}$	77.8	$dt_{4,2,2}$	10.0
$A_{4,1,2}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,2,3}$	106.1
$A_{4,2,1}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,3,3}$	183.3
$A_{4,2,2}$	0.0	$dt_{1,5,2}$	66.7	$dt_{4,4,1}$	122.2

Variable	Value	Variable	Value	Variable	Value
$dt_{4,4,2}$	10.0	$f_{1,5,2}^H$	0.0	$f_{5,4,2}^H$	15.63615505807977
$dt_{4,4,3}$	94.5	$f_{2,1,2}^H$	10.6	$f_{5,5,1}^H$	0.0
$dt_{4,5,1}$	122.2	$f_{2,2,2}^H$	0.0	$f_{5,5,2}^H$	0.0
$dt_{4,5,2}$	10.0	$f_{2,3,1}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,3,2}^H$	0.0	$f_{1,2,1}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,4,1}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,2}$	10.0	$f_{2,4,2}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,1,3}$	138.9	$f_{2,5,1}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,2}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,2}$	74.66004781348396	$f_{3,1,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,3}$	74.66004781348391	$f_{3,1,2}^H$	13.98017677462762	$f_{1,5,1}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	5.214090626939798	$f_{1,5,2}^C$	0.0
$dt_{5,3,2}$	10.0	$f_{3,2,2}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,4,3}$	116.7	$f_{3,3,2}^H$	0.8198232253723838	$f_{2,3,1}^C$	0.0
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,2}$	105.6	$f_{3,4,2}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,2}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,1}^H$	0.0	$f_{2,5,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	12.6	$f_{2,5,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,1}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,2,2}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,3,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,3,1}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,1}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,4,2}^C$	17.3
$f_{1,3,1}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{3,5,2}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,2,2}^H$	2.0638449419202285	$f_{4,1,2}^C$	5.304500000000011
$f_{1,4,2}^H$	0.0	$f_{5,3,1}^H$	5.4355588897224285	$f_{4,2,1}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,2,2}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,3,2}^C$	8.4	$\text{RecLMTD}_{3,2,1}$	0.026450800778310225	$d_{1,2,1}$	0.0
$f_{4,4,1}^C$	0.0	$\text{RecLMTD}_{3,2,2}$	0.01209361792597359	$d_{1,2,2}$	0.0
$f_{4,4,2}^C$	0.0	$\text{RecLMTD}_{3,3,1}$	0.014388920828156229	$d_{1,3,1}$	0.0
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,3,2}$	0.02150212335787724	$d_{1,3,2}$	0.0
$f_{4,5,2}^C$	13.9	$\text{RecLMTD}_{3,4,1}$	0.029297885186065308	$d_{1,4,1}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.0064938037725891535	$d_{1,4,2}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{3,5,2}$	0.022306518908293102	$d_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.024030191698075007	$d_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.02554670259920889	$d_{2,1,2}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.009469696969696968	$d_{2,2,2}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.009423141115769086	$d_{2,3,1}$	0.0
$f_{5,4,2}^C$	0.0	$\text{RecLMTD}_{4,3,2}$	0.008535688158765427	$d_{2,3,2}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.012387051800747543	$d_{2,4,1}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.045618305265862594	$d_{2,4,2}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,1}$	0.04305097282506776	$d_{2,5,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.015434474530129737	$d_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.024141568940966463	$d_{3,1,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.1	$d_{3,1,2}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.1	$d_{3,2,1}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,2,2}$	0.1	$d_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.012853470437017995	$\text{RecLMTD}_{5,3,1}$	0.1	$d_{3,3,1}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.028888218232147534	$d_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,4,2}$	0.1	$d_{3,4,1}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.1	$d_{3,4,2}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.1	$\text{RecLMTD}_{5,5,2}$	0.009469696969696968	$d_{3,5,2}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115769086	$d_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.020926522199679467	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$d_{4,1,2}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.1	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$d_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.01551425892990688	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$d_{4,2,2}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$d_{4,3,2}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.1	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$d_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$d_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$q_{1,1,1}$	0.1	$d_{4,5,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{4,5,2}$	0.0	$t_{2,5,2}^H$	198.85754716980975	$t_{1,4,1}^C$	102.85953757225435
$q_{5,1,1}$	0.0	$t_{3,1,1}^H$	175.34864864864863	$t_{1,4,2}^C$	82.2
$q_{5,1,2}$	0.0	$t_{3,1,2}^H$	113.43219448598055	$t_{1,5,1}^C$	204.4
$q_{5,2,1}$	0.0	$t_{3,2,1}^H$	226.7	$t_{1,5,2}^C$	93.3
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	65.6	$t_{2,1,2}^C$	60.0
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,2}^C$	115.6
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	74.52388659510073	$t_{2,3,1}^C$	37.8
$q_{5,4,2}$	0.0	$t_{3,4,1}^H$	172.99324324324346	$t_{2,3,2}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	106.84594594595175	$t_{2,4,1}^C$	147.35163152253665
$q_{5,5,2}$	0.0	$t_{3,5,2}^H$	65.6	$t_{2,4,2}^C$	112.86184971098264
$q_{cu,2}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,1}^C$	93.3
$q_{cu,4}$	0.0	$t_{4,1,2}^H$	148.9	$t_{2,5,2}^C$	148.34264863679172
$q_{hu,1}$	0.0	$t_{4,2,1}^H$	237.58333333333334	$t_{3,1,1}^C$	127.37876463650335
$q_{hu,2}$	0.0	$t_{4,2,2}^H$	148.9	$t_{3,1,2}^C$	60.0
$q_{hu,3}$	0.0	$t_{4,3,2}^H$	148.9	$t_{3,2,1}^C$	115.6
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	148.9	$t_{3,3,1}^C$	111.5
$t_{1,1,1}^H$	160.0	$t_{4,5,1}^H$	271.1	$t_{3,3,2}^C$	37.8
$t_{1,2,1}^H$	160.0	$t_{4,5,2}^H$	148.9	$t_{3,4,1}^C$	104.7672291040321
$t_{1,2,2}^H$	160.0	$t_{5,1,1}^H$	198.9	$t_{3,4,2}^C$	82.2
$t_{1,3,1}^H$	160.0	$t_{5,1,2}^H$	65.6	$t_{3,5,2}^C$	166.23812949640183
$t_{1,3,2}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	60.00012385980429
$t_{1,4,1}^H$	160.0	$t_{5,2,2}^H$	108.74827200025184	$t_{4,1,2}^C$	60.0
$t_{1,4,2}^H$	119.38522727272726	$t_{5,3,1}^H$	198.9	$t_{4,2,1}^C$	221.7
$t_{1,5,1}^H$	160.0	$t_{5,3,2}^H$	65.6	$t_{4,2,2}^C$	115.6
$t_{1,5,2}^H$	160.0	$t_{5,4,2}^H$	95.99923955901362	$t_{4,3,2}^C$	37.8
$t_{2,1,2}^H$	137.8	$t_{5,5,1}^H$	198.9	$t_{4,4,1}^C$	104.7672291040321
$t_{2,2,2}^H$	137.8	$t_{5,5,2}^H$	65.6	$t_{4,4,2}^C$	82.2
$t_{2,3,1}^H$	248.9	$t_{1,1,1}^C$	90.20407122822533	$t_{4,5,1}^C$	204.4
$t_{2,3,2}^H$	137.8	$t_{1,2,1}^C$	115.6	$t_{4,5,2}^C$	93.3
$t_{2,4,1}^H$	241.61331822282358	$t_{1,2,2}^C$	115.6	$t_{5,1,1}^C$	92.62148308310525
$t_{2,4,2}^H$	137.8	$t_{1,3,1}^C$	111.5	$t_{5,1,2}^C$	90.20394736842104
$t_{2,5,1}^H$	198.85754716981077	$t_{1,3,2}^C$	107.67619047619047	$t_{5,2,1}^C$	115.6

Variable	Value	Variable	Value	Variable	Value
$t_{5,2,2}^C$	115.6	$b_{4,1,2}^{H,in}$	1876.14	$b_{2,5,2}^{H,out}$	0.0
$t_{5,3,1}^C$	111.5	$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0
$t_{5,3,2}^C$	37.8	$b_{4,2,2}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	1625.0160748080839
$t_{5,4,2}^C$	146.03815028901732	$b_{4,3,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	1182.0343451272522
$t_{5,5,1}^C$	204.4	$b_{4,4,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0
$t_{5,5,2}^C$	124.01079136690645	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	185.85392519191942
$b_{1,2,1}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	0.0	$b_{3,5,2}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	321.8816879954265	$b_{4,1,2}^{H,out}$	1876.14
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	1081.132663165791	$b_{4,2,1}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,4,2}^{H,in}$	1563.7983120045737	$b_{4,3,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	1460.6799999999999	$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	321.8816879954265
$b_{3,1,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	1081.1326631657912
$b_{3,1,2}^{H,in}$	1625.0160748080839	$b_{1,5,1}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	1182.0343451272522	$b_{1,5,2}^{H,out}$	0.0	$b_{5,4,2}^{H,out}$	1563.7983120045737
$b_{3,2,2}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	1460.6799999999999	$b_{5,5,1}^{H,out}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	0.0
$b_{3,3,2}^{H,in}$	185.85392519191942	$b_{2,3,1}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{1,4,1}^{C,in}$	0.0	$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	318.27000000000066
$b_{1,4,2}^{C,in}$	0.0	$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0
$b_{1,5,1}^{C,in}$	0.0	$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0
$b_{1,5,2}^{C,in}$	0.0	$b_{5,4,2}^{C,in}$	0.0	$b_{4,3,2}^{C,out}$	317.52
$b_{2,1,2}^{C,in}$	0.0	$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0
$b_{2,2,2}^{C,in}$	705.16	$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0
$b_{2,3,1}^{C,in}$	0.0	$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	1296.87
$b_{2,4,1}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,4,2}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	705.16	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	1422.0600000000002	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	318.27000000000066	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,3,2}^{C,in}$	317.52	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	1296.87	$b_{3,4,1}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	1422.0600000000002		
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		